

Youngmin Jo

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Education

3/2024 – 8/2027 B.S. in Mechanical Engineering, **Sogang University**
GPA: 3.88/4.0
Summa Cum Laude (expected), pending final semester GPA

Research Experience

12/2025 – present **Undergraduate Researcher**
Materials Design and Process Engineering Laboratory, **Sogang University**
Advisor: Prof. Heonjae Jeong

2/2026 – 8/2026 **Origin of Negative Photoconductivity in Amorphous ZnON**
• Performed first-principles calculations to investigate the origin of negative photoconductivity (NPC) in amorphous ZnON
• Identified charge localization mechanisms at the valence band maximum and conduction band minimum through AIMD and hybrid DFT simulations

7/2026 – present **Thermodynamic Stability and Li-Ion Conductivity of Doped LLZO Solid-Electrolyte Interfaces**
• Evaluating interfacial thermodynamic stability and Li-ion conductivity of Al-, Ga-, Ta-, and Sr+Ta-doped LLZO solid electrolytes at LiSi anode and FeF₃ cathode interfaces for thermal-battery applications
• Assessing dopant-dependent interfacial phase stability and Li-ion transport under realistic operating conditions (500 °C) using first-principles calculations

Publications

Co-author (2nd author), "Origin of Negative Photoconductivity in Amorphous ZnON" (working title), manuscript in preparation, expected 2026.

Technical Projects

3/2026 – present **Automated Robotic Arm Teleoperation System**
Developed a motion-driven robotic arm system capable of real-time teleoperation to automate repetitive lab experiments and manual labor tasks. Streamlined the motor control and teleoperation pipeline by integrating the LeRobot package, replacing traditional hard-coding methods and enhancing remote-control accessibility.
Skills: LeRobot package, Computer vision, Linux, Autodesk Inventor, 3D printing

8/2025 – 12/2025 **Predictive Weather Navigation System**
Developed a dynamic route-planning application in C that synchronizes spatial coordinates from a Navigation API with temporal data from a Meteorological API to predict weather conditions at specific waypoints along a route. Engineered a data processing pipeline to extract, parse, and integrate raw external datasets into a functional format, building data manipulation and integration skills directly applicable to handling large-scale datasets in computational research.
Skills: C, REST APIs (Navigation & Weather)

Skills

Computational Materials Design: VASP, Gaussian, Pymatgen, ASE, VESTA, Avogadro

Programming & Systems: Python, C, Linux/Bash

Engineering & Design: Arduino, MATLAB, AutoCAD, Autodesk Inventor, 3D Printing

Honors & Awards

Fall/2026 (expected)	President's Award & Dean's List with Highest Honors (Top 1%), Sogang University
Summer/2026	President's Award & Dean's List with Highest Honors (Top 1%), Sogang University
10/2025	Encouragement Prize, 11th DB GAPS Investment Competition, DB Financial Investment
8/2025	Best Team Award, DS Division, Samsung Electronics, South Korea
8/2025 – 12/2027 (ongoing)	Ilju Academy and Culture Foundation Scholarship, Taekwang Group
3/2025	Alumni Association Scholarship

Extracurricular Activities

3/2024 – 6/2027 (ongoing)	President (Former: Vice President, Executive Committee Member) Mechanical Engineering Society Directed member engineering projects and managed a technical corporate collaboration with Taejin Technology (HTC KOREA)
9/2025 – present	English Tutor (Volunteer) Local Children's Center, South Korea Implemented English lesson plans and academic mentorship to five middle school students on a bi-weekly basis
8/2025	Participant Samsung Shining Star 2025, Samsung Electronics Completed an intensive training program focused on semiconductor memory manufacturing and the 8-step fabrication process
12/2024 – 2/2025	Global Volunteer (Exchange Participant) AIESEC in FHCMC, Vietnam Taught English to secondary students and facilitated Korean language and cultural exchange sessions for university students during 210 hours of service.